Variation of Height Pairings

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Motivation

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Theorem (Silverman)

Let A/S be an abelian scheme. For every sections $P \in A(S), Q \in A^\vee(S)$ and any Weil height $h_{\bar{S}}$ associated to a degree-one line bundle on $\bar{S}$, one has

$$\lim_{s \in S(\bar{K}), h_{\bar{S}}(s) \to \infty} \frac{\langle P_s, Q_s \rangle}{h_{\bar{S}}(s)} = \langle P_\eta, Q_\eta \rangle \in \mathbb{Q}.$$

(RHS stands for the geometric Néron–Tate height pairing.)

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Theorem (C.'26)

There exists an integrable arithmetic $\mathbb{Q}$-adelic line bundle $\widehat{M}$ on $\bar{S}$, depending on P, Q, such that the height function $h_{\widehat{M}}$ satisfies

$$h_{\widehat{M}}(s) = \langle P_s, Q_s \rangle, \quad s \in S(\bar{K})$$

(without error terms).

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- This recovers Silverman's theorem.

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$$h_{\widehat{M}}(s) = (-1)^p \langle Z_s, W_s \rangle, \quad s \in S(\bar{K}).$$

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Moreover, if $Z = W$, the adelic line bundle $\widehat{M}$ is nef. In particular,

$$(-1)^p \langle Z_s, Z_s \rangle \geq 0,$$

where $p \coloneqq \operatorname{codim}_{X_s} Z_s$.

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where $h_{\bar{S}}$ is any Weil height associated to a degree-one line bundle on $\bar{S}$ and the RHS is the geometric height pairing.

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- The geometric height pairing should be given by the degree of the geometric part of $\widehat{M}$.
- Variation of the local archimedean height pairing for algebraically trivial cycles is studied in our work (2025).

Thank you for your attention!